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Multi-branch AC arc fault detection based on ICEEMDAN and LightGBM algorithm

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ABSTRACT

The operation characteristics of household appliances would interfere with the arc fault detection. Especially in the case of the multi-branch circuit, the arc fault information is mixed with the operation characteristics of various loads, which makes the traditional arc fault detection algorithm invalid. In this paper, arc fault signals are acquired from the designed multi-branch experimental platform composed of masking loads according to IEC 62606. By adding the adaptive noise, ICEEMDAN algorithm is found to be suitable to extract more sufficient arc fault information. By selecting the IMF and constructing its variance, the detection variables are constructed to distinguish the arc fault from the normal state. Compared with wavelet transform, the ability to extract arc fault information is improved by 8.47 times on average. In order to simplify the algorithm structure, the Boruta and LightGBM algorithm are used to reduce the feature dimension and improve the algorithm efficiency. Based on the well-trained LightGBM model, the arc fault detection algorithm is finally proposed, which could detect multi-branch arc faults accurately and fast. It is proved to have the higher detection accuracy of 97.06% and the lower detection time of no more than 300 ms on the Raspberry pi platform.

1. Introduction

Arc faults caused by insulation failure or loose connection could pose a great threat to line safety. With the increase of the system capacity, the increasing current level makes arc faults more likely to lead to electrical fire accidents [1]. In order to detect arc faults in time and reduce their harm as soon as possible, the USA issued the UL1699 requiring the installation of arc fault circuit interrupters (AFCI) in all living areas [2]. The IEC 62606 standard in Europe has also designed a relatively complete arc fault test to detect the effectiveness of the device's detection performances [3].

Arc faults mainly consist of series arc faults and parallel arc faults according to the arc fault location. Parallel arc faults could also be divided into phase-to-phase arc faults and phase-to-ground arc faults [4, 5]. As for parallel arc faults, the arc fault current is much higher than the normal current, which is easy to be protected by the overcurrent detection function of interrupters [6]. The current of series arc faults does not change significantly and the series arc fault detection is more likely to be interfered by loads in series [7]. The operation characteristics of household appliances would inhibit the arc fault characteristics during the detection process, so they are also defined as masking loads in the IEC 62606 standard. Therefore, its potential harm is much greater, which has attracted more attention from arc fault detection researches [8].

The location of the arc fault cannot be judged in advance, and the analysis based on the arc fault current could solve the problem caused by the randomness [9,10]. After the arc fault occurs, the current would be distorted, and the zero-crossing time is the most representative period. By calculating the variance, integral and fractal of the current, the arc fault is distinguished from the normal state in time domain [11,12,13]. Current fluctuation features and the zero-current feature is proposed to obtain the comprehensive feature identification coefficient of the series arc faults [14]. The frequency estimation of electrical signals could also help detect the arc fault in power systems [15]. Besides, the arc fault also increases the high-frequency harmonic components of the current, so the time-frequency analysis method has become a commonly method for the feature extraction. Time-frequency analysis methods such as fast Fourier transform, short-time Fourier transform and wavelet transform could effectively extract the change of the current frequency component with time, which could be used to determine the high-frequency

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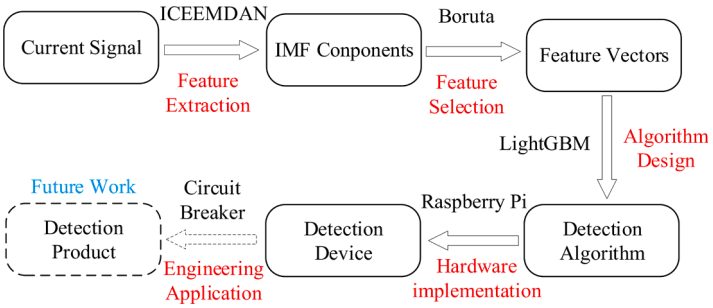


Fig. 1. Research flowchart of the paper.

characteristics of the arc fault [16,17,18]. The differential equation algorithm (DEA) could analyze the harmonic signals of current and voltage to calculate the line parameters, which helps detect and locate the arc fault [19,20]. Empirical mode decomposition (EMD) could decompose the arc fault into natural mode signals, and its instantaneous amplitude could be constructed as the arc fault detection feature [21]. Ensemble empirical mode decomposition (EEMD) is used to help differential protection in AC microgrid [22].

Based on the constructed detection variables, the design methods of detection algorithms are also various. The threshold comparison algorithm is simple in logic and fast in operation, which is verified to achieve a detection accuracy of 90.91% [23]. With the further development of machine learning, intelligent feature fusion has been shown to better adapt to the arc fault detection requirements in different scenarios [24]. Artificial neural network (ANN) is applied to discover the complicated behavior and regularity in constructed features, whose detection accuracy achieves 99.21% [25]. Support vector machine (SVM) is used to deal with linear inseparable classification problems for arc fault detection, and the detection accuracy could reach 95.5% [26]. The convolutional neural network (CNN) model is used to construct the arc fault detection algorithm by recognizing the image based on the arc fault feature [27].

Existing researches on arc fault detection for different single loads have acceptable results. However, most of them neglect the influence of multiple branches, which is more common in reality. In this case, the arc fault detection accuracy might be affected when the arc fault occurs in different positions such as bus or branch. It is necessary to study the arc fault characteristics in the multi-branch circuit. Besides, the detection algorithm based on machine learning methods would occupy a large amount of memory, and the calculation speed is slow, which affects the arc faults detection time. It puts forward more stringent requirements for the detection capability of arc fault detection algorithm.

This paper provides an effective algorithm to detect multi-branch arc faults for various masking loads. Contributions of the paper could be summarized as follows:

- An experimental platform for multi-branch arc fault detection is designed according to IEC 62606. The ICEEMDAN algorithm is found to be suitable for different scenarios. It helps to construct detection variables to accurately distinguish arc faults from normal states.
- For multi-branch circuits composed of resistors and household loads, two evaluation indexes are constructed to evaluate the effectiveness of arc fault detection variable. Based on this, the interference of the arc fault location and household appliances on arc fault detection is analyzed.
- It is difficult for machine learning methods to meet the requirements of fast detection in practical applications due to the constraints of detection cost. This paper uses the Boruta algorithm as a feature selecting method to reduce the data dimension and applies the LightGBM algorithm to simplify the training model structure. The proposed algorithm is implemented on the Raspberry pi platform, which has satisfying detection performance.

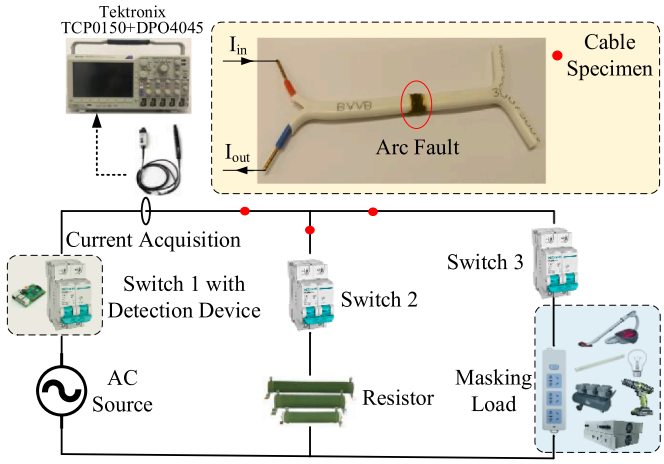


Fig. 2. The diagram of multi-branch arc fault experimental platform with different kinds of masking loads.

The rest of the paper is organized as shown in Fig. 1: a multi-branch arc fault experimental platform is built and the current waveform of the bus is measured. The interference of masking loads on arc fault detection characteristics is analyzed by time-domain and time-frequency domain methods. The detection efficiency is improved through the feature selection and the algorithm design. Finally, the proposed algorithm is verified on the hardware platform. The future work is to combine the involved algorithms with circuit breakers and install them into the actual power system.

2. Discussion on the effectiveness of existing arc fault detection variables for masking loads

Based on the designed experimental platform according to the IEC 62606 standard, resistors and different types of masking loads are connected to the circuit in parallel. By changing the arc fault position, the interference of masking loads on arc fault detection is analyzed. The proposed features effectively improve the ability of arc fault identification.

2.1. Experimental Platform and method

As shown in Fig. 2, different types of loads are connected to the system circuit and the resistance is connected in parallel. The operation sequence of loads could be determined by the switches.

The cable specimen is used to ignite the arc fault, which is composed of two originally insulated cable conductors. Conductors are cut longitudinally to destroy their insulation, then a carbonized conductive path between the two cable conductors is formed by a 7 kV power supply [3]. The current in the carbonization process is limited to no more than 30 mA. The cable specimen could be connected to the position set in the

Table 1
Different types of masking loads and their parameters.

Load type	Masking load	Parameters
Electric energy - Kinetic energy	Electric drill	600 W
	Capacitor start (air compressor type) motor	2200 W
	Vacuum cleaner	1150 – 1610 W
Electric energy - Thermal energy	Resistor	575 – 4600 W
	Electronic lamp dimmer (Thyristor type) *	600 W
	Halogen lamp	At least 300 W
Electric energy - Light energy	Fluorescent lamp	40 W
Power related	EMI fiter	0.22 μF
	Electronic switching mode power supply	Minimum total harmonic distortion (THD) of 100%, individual minimum current harmonics of 75% at the 3rd, 50% at the 5th, and 25% at the 7th

* The dimmer is to be turned on with the dimmer preset at full on, conduction angles of 60°, 90°, 120°, and at the minimum setting that causes the lamps to ignite.

circuit to generate an arc fault. The arc fault current is sampled by Tektronix TCP0150 and recorded by Tektronix DPO4045 Memory Recorder at the sampling frequency of 1 MHz.

Compared with the traditional pull-apart method, the cable specimen is more suitable for AC arc fault. It could generate arc fault at a lower current level while maintaining the self-sustaining state. AC arc fault currents have zero-crossing point, making the pull-apart method easy to cut off the circuit, which leads to the arc fault generation difficulty. The cable specimen needs to be replaced to maintain the effective generation of arc faults after several times of use. Since the current of the resistor is sinusoidal in time domain and only has white noise interference in frequency domain, the interference of the resistor is relatively limited compared with masking loads. It is suitable for the resistor to be installed in another branch to analyze the interference of the single masking load.

There are various types of household appliances, and it is necessary to select representative loads for arc fault testing to verify the universality of detection. Table 1 lists 9 different types of household appliances (defined as masking loads in IEC 62606) and their parameters, basically covering most household load types [3].

According to the IEC 62606 standard, by changing the state of switches and the position of cable specimen, the experimental platform could be designed into four types of configurations shown in Fig. 3. The red dot indicates the arc fault location and the green circle indicates the current sensor location. The system configuration A is used to research the arc fault detection of single branch circuit, as a comparison to analyze the interference of multi-branch conditions. The other three system configurations are all used to research the arc fault detection of multi-branch circuits. Masking loads are used to form a branch. In order to better analyze the impact of masking loads on the arc fault detection, another branch uses resistor as load, which would not introduce too complex additional interference.

2.2. Limitation Analysis of detection variable based on wavelet transform

Different types of masking loads in multi-branch circuit would interfere with arc fault detection. Therefore, extracting effective arc fault detection variables as much as possible has become the most important research content of arc fault detection algorithm.

Because of the instantaneity and satisfying discriminant effect of time-frequency analysis, it is considered to be one of the most suitable methods for arc fault detection. Compared with FFT and short-time Fourier transform, wavelet transform could provide an analysis window changing with the frequency, so that detection variable could obtain both suitable time resolution and frequency resolution [28,29]. It could be defined as Eq. (1):

$$I(t) = \sum_{k=-\infty}^{\infty} \sum_{n=\infty}^{\infty} d(k,n)\psi(k,n,t), d\&\psi \in L^2(R) \quad (1)$$

where $I(t)$ is the measured current signal, $d(k,n)$ is the two-dimensional coefficients obtained by decomposition, $\psi(k,n,t)$ is the transformation basis, k and n are two mapping axes respectively. The Rbio3.1 mother wavelet could clearly reflect the arc fault characteristics because of its

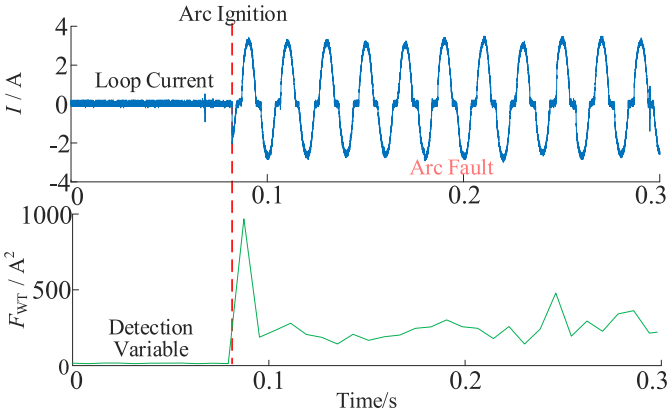


Fig. 4. Arc fault current alone of the resistor and its detection variable based on the wavelet transform (Single branch).

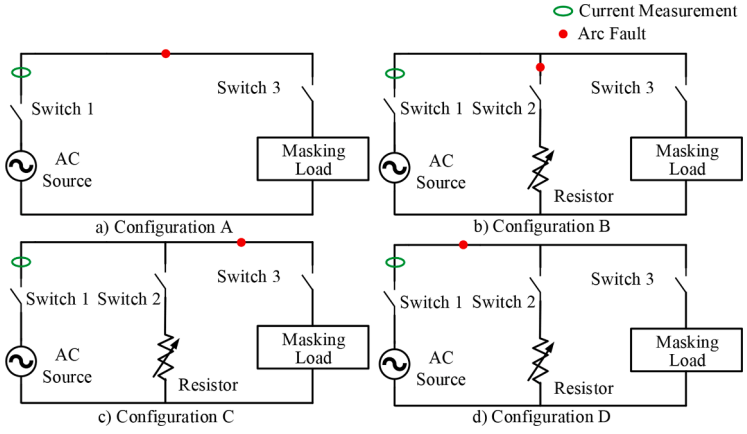


Fig. 3. Different system configurations of multi-branch arc fault experiment.

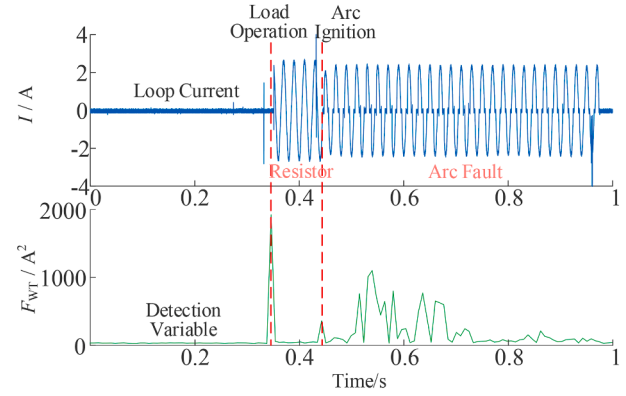


Fig. 5. Arc fault current of the resistor and its detection variable based on the wavelet transform (Single branch).

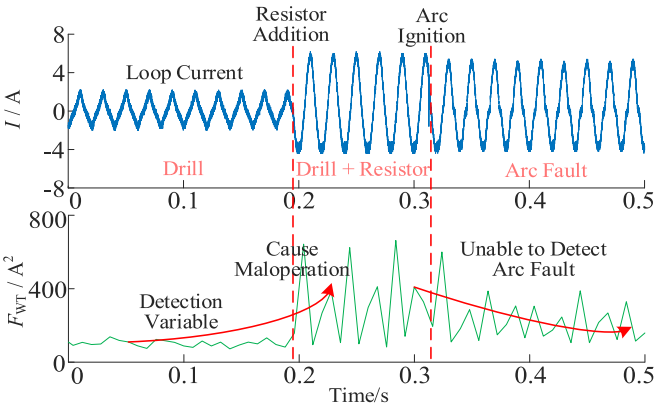


Fig. 6. Arc fault current of the electric drill and its detection variable based on the wavelet transform (multi-branch).

suitable vanishing moment, symmetry and biorthogonality [30]. After choosing 15.6 k-23.4 kHz as the proper frequency band and applying 6-level decomposition, the detection variable $F_{wt}(t)$ is calculated shown in Eq. (2) to display the differences between arc faults and normal states:

$$F_{wt}(tn) = \sum_{t=t_0}^{tn} d^2(j,t) \tag{2}$$

where t_0 and t_n is the start and end time of the analysis time window separately, the analysis time window length is 8 ms, j is selected frequency band number.

In case of inserting a resistor with series arc fault shown in Fig. 4, the detection variable based on wavelet transform increases after the arc fault occurs, which could distinguish the arc fault from the normal state. Fig. 5 shows the arc fault waveform after the resistor runs normally for about 0.1 s. There is no obvious amplitude change of the detection variable after the resistor work and there is only a pulse in the time window of current transient. The detection variable amplitude does not change until the arc fault occurs. The detection variable based on wavelet transform could distinguish the current change caused by normal operation of loads and arc faults clearly, so that it would not easily cause maloperation.

When the masking load and resistor are connected to the circuit, the Gaussian noise of resistor and the interference of masking load are superimposed into the current signal, which makes the detection variable based on wavelet transform no longer effective. As shown in Fig. 6, after the resistor is turned on, the amplitude of the detected variable increases and fluctuates significantly, which is different from that when the electric drill works alone. It is easy to lead to too many false positive results. After the arc ignition, there is no significant difference of the detection variable between the arc fault period and the previous normal period.

It is hard for the detection variable based on wavelet transform to effectively distinguish the arc fault from the normal state. Therefore, it is urgent to propose new feature construction methods to extract obvious arc fault characteristics.

3. Research on optimal feature construction for multi-branch arc fault detection

It could be seen that the detection variables based on wavelet transform have satisfying performance in detecting arc faults for the single load circuit, but cannot detect arc faults for multi-branch circuit. Feature extraction methods, such as wavelet transform, do not have adaptive capability, whose parameters cannot be changed according to different environments, making it only effective for some detection environments. An adaptive feature extraction method could extract more effective arc fault information for different types of masking loads and different arc fault locations.

3.1. Construction and evaluation of effective arc fault detection variables

EMD is usually used to decompose the arc fault signal according to the time scale characteristics of the signal itself. Compared with the wavelet analysis, its adaptive ability helps it have advantages for analysis of nonlinear and non-stationary time series signal. EEMD solves the problem of mode aliasing in practical application of EMD by introducing Gaussian noise, but the residual noise interferes with further analysis and processing.

Torres ME et al. proposed Complete EEMD with adaptive noise (CEEMDAN) and improved CEEMDAN (ICEEMDAN) algorithms by improving the decomposition and noise introduction process [31,32]. In the decomposition process, intrinsic mode function (IMF) components of white noise decomposed by EMD are introduced into IMF components of the signal, effectively reducing the residual noise. Besides, the local mean is used to replace the modal estimation to solve the problem of white noise transfer with the decomposition process. $x(t)$, $\langle\cdot\rangle$, $\eta^i(t)$ are respectively the original arc fault signal, the average operation and the Gaussian noise with zero mean. i represents the number of noises added and β_k is used to remove noise. The EMD algorithm with operators is $E_k(\cdot)$ and $M(\cdot)$ is used to calculate the local mean of the signal. The ICEEMDAN could be defined as follows:

Step1: Construct signal with the noise and calculate the first residual $\gamma_1(t)$:

$$x^i(t) = x(t) + \beta_0 E_1\left(\eta^i(t)\right) \tag{3}$$

$$\gamma_1(t) = \left\langle M\left(x^i(t)\right)\right\rangle \tag{4}$$

Step2: Obtain the first order component of IMF:

$$IMF_1(t) = x(t) - \gamma_1(t) \tag{5}$$

Step3: Calculate the second order component of IMF by using the local average:

$$\gamma_2(t) = \left\langle M\left(\gamma_1(t) + \beta_1 E_2\left(\eta^i(t)\right)\right)\right\rangle \tag{6}$$

$$IMF_2(t) = \gamma_1(t) - \gamma_2(t) = \gamma_1(t) - \left\langle M\left(\gamma_1(t) + \beta_1 E_2\left(\eta^i(t)\right)\right)\right\rangle \tag{7}$$

Step4: Calculate other order components of IMF and residuals:

$$\gamma_k(t) = \left\langle M\left(\gamma_{k-1}(t) + \beta_{k-1} E_k\left(\eta^i(t)\right)\right)\right\rangle \tag{8}$$

$$IMF_k(t) = \gamma_{k-1}(t) - \gamma_k(t), k = 3, 4, \ldots, N \tag{9}$$

As shown in Fig. 7, ICEEMDAN is used to decompose the arc fault

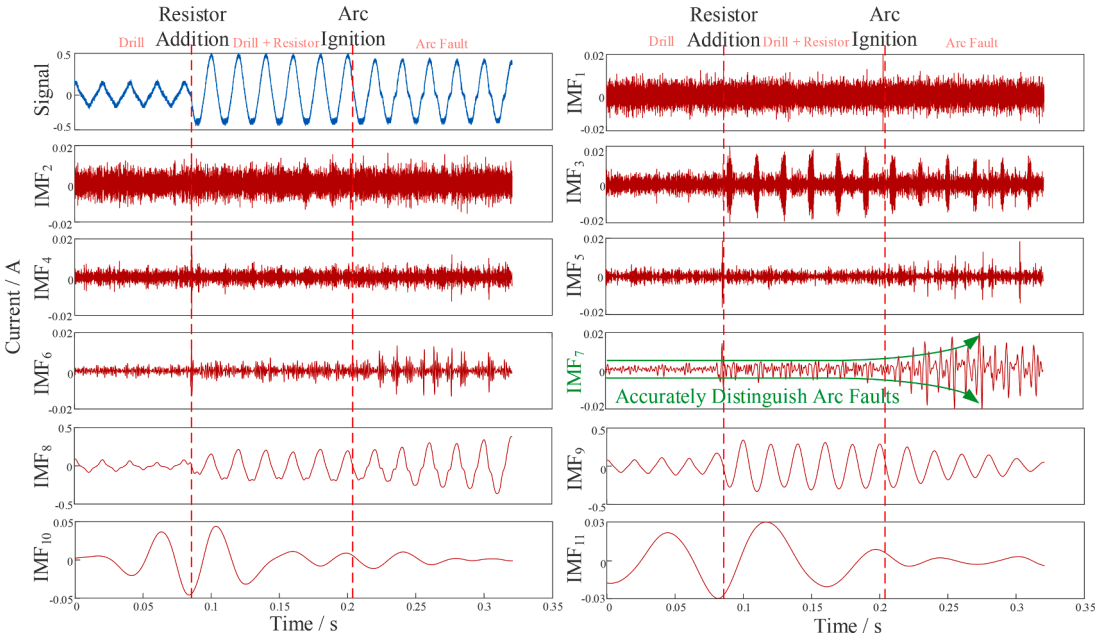


Fig. 7. 1–11 order IMF components of arc fault waveform obtained by ICEEMDAN.

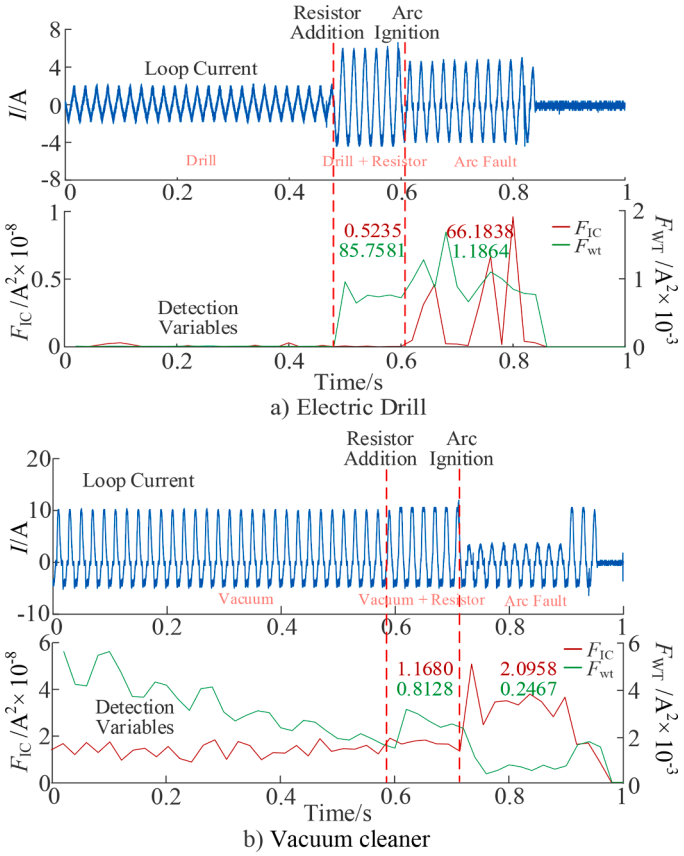


Fig. 8. Arc fault detection variables of typical load based on the wavelet transform and the ICEEMDAN algorithm (multi-branch).

waveform. IMF6 and IMF7 have large amplitude fluctuation after the arc fault occurs, which are suitable to extract effective information to distinguish arc faults from the normal state. In order to complete arc fault detection in a short time, the time window is necessary to be added to the analysis signal like wavelet transform. Amplitude fluctuations are quantitatively calculated in the form of variances, from which the detection variable $F_{IC}(t)$ is calculated shown in Eq. (10):

$$F_{IC}(tn) = \frac{\sum_{t=t_0}^{tn} \left(IMF_{k,t} - \overline{IMF_k}\right)^2}{t_n - t_0} \tag{10}$$

where the analysis time window length is 20 ms, k is the selected IMF component number. Besides, in order to verify the applicability of the ICEEMDAN algorithm for different masking loads, two indexes are designed as following to quantitatively describe the effectiveness of detection variables.

$$Q_{1,\psi} = \text{mean}\left(F_{\psi,R}\right)/\text{mean}\left(F_{\psi,L}\right) \tag{11}$$

$$Q_{2,\psi} = \text{mean}\left(F_{\psi,Arc}\right)/\text{mean}\left(F_{\psi,R}\right) \tag{12}$$

where ψ is the selected method, which could be the wavelet transform and the ICEEMDAN algorithm. $F_{\psi,R}$ and $F_{\psi,A}$ are detection variable values without the large pulse in resistor addition periods and arc faults periods, $F_{\psi,L}$ is all detection variable values during the masking load operating alone, mean is the operator to obtain the average value. It is necessary to make the index $Q_{2,\psi}$ as large as possible and the index $Q_{1,\psi}$ as small as possible.

Fig. 8 shows the arc fault detection variables of typical masking loads. Two indexes are also calculated and marked. As for the electric drill, the detection variable based on the wavelet transform has a significant increase due to the added resistor and $Q_{1,WT}$ is 85.7581, so it is easy to lead to maloperation. After the arc fault occurs, the detection variable nearly does not change and $Q_{2,WT}$ is only 1.1864, which means it lacks ability to extract arc fault information effectively. The ICEEMDAN-based detection variable still keeps an accurate distinction between arc faults and the normal state. $Q_{1,IC}$ and $Q_{2,IC}$ are 0.5235 and 66.1838, satisfying the detection demand. For the vacuum cleaner, the detection variable based on wavelet transform shows a decreasing trend after the resistor is connected and arc fault occurs, which is significantly different from other masking load conditions. $Q_{2,WT}$ is 0.2467, which means it isn't applicable for arc fault detection at all. The detection variable based on IMF7 of ICEEMDAN has little change in amplitude after the resistor adding process. It shows an increasing amplitude pattern immediately after the arc fault occurrence. $Q_{2,IC}$ is 2.0958,

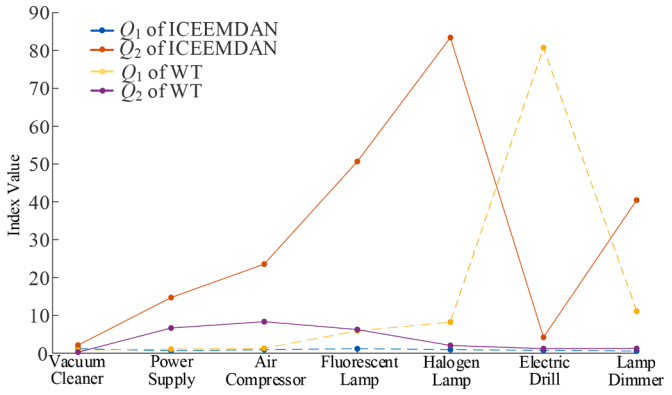


Fig. 9. Comparison of two indexes with different detection variables for different masking loads (multi-branch).

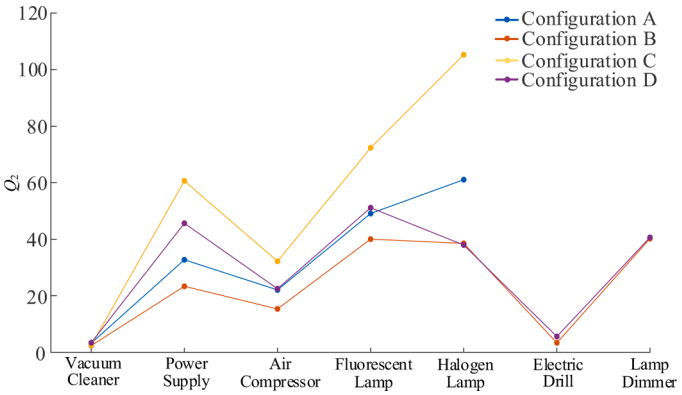


Fig. 10. Comparison of the $Q_{2,IC}$ index for different system configurations.

proving the detection variable has a accurate detection performance.

Through multi-branch arc fault experiments for different types of masking loads, the average index values of different detection variables are calculated as shown in Fig. 9. During the resistor addition period, the yellow line shows that the wavelet-based detection variable performs well for the vacuum cleaner, power supply and air compressor. For masking loads of electrothermal and electro-optical types, the wavelet-based detection variable is susceptible to interference by resistor, and finally results in malfunction in the normal state. Besides, as shown by the purple line, $Q_{2,WT}$ is significantly smaller than $Q_{1,WT}$, which reflects that the wavelet-based detection variable could not effectively distinguish the resistor addition process from the arc fault. For ICEEMDAN-based detection variable, the two processes could be distinguished accurately, and it is superior to wavelet-based detection variable regardless of masking load types.

As a time-frequency analysis, wavelet transform could only extract the time-frequency information of current according to the selected frequency band. The operation characteristics of different types of masking loads are quite different, so that it is difficult to construct universally valid detection variables through a single detection frequency band. It is also hard to avoid the possible maloperation caused by the resistor addition process, which changes the time-frequency information of the loop current. ICEEMDAN decomposes the analyzed signal into monotonous residual components (not shown in Fig. 7) by adaptively adding noise for modal decomposition, which helps to obtain more detailed information of the signal change process. The detection variables obtained by IMF_3 in Fig. 7 and wavelet transform are similar. Although they could capture the difference of current change, they cannot distinguish between the normal state and the arc fault. The IMF_3 is decomposed three times again, which finally filters the current change information caused by the normal operation. Compared with the wavelet transform, ICEEMDAN could extract more detailed information, effectively construct arc fault detection variables. Its adaptive ability helps it adapt to more masking load types.

3.2. Influence analysis of different configurations on arc fault detection variables

Due to the masking load branch and the resistor branch as shown in Fig. 2, the arc fault at different locations would have different effects on the current signals measured at the power supply side. The operating current of the electric drill and the lamp dimmer is small, so that when the arc fault occurs in its branch it is difficult to form a continuous arc fault state. Only system configurations B and D are considered for these two types of masking loads.

Q_2 is more important than Q_1, which reflects the sensitivity of detection variables to the arc fault information. In most cases, the larger Q_2 value is based on the smaller Q_1 value. Fig. 10 shows the $Q_{2,IC}$ index for different system configurations. The system configuration B, C, D considers the multi-branch arc faults, which are different from the system configuration A in the experimental process. Therefore, the detection variable performance of the three system configurations are mainly compared.

For the same type of masking loads, the Q_2 value of system configuration B is always the minimum while the Q_2 value of system configuration C is the maximum. When the arc fault occurs in the masking load branch, the arc fault noise is mixed with the operation characteristics of the masking load, which makes the current characteristics change greatly. In addition, the resistor in parallel mainly provides Gaussian noise, which is similar to most environmental interference noises and has limited interference to the bus current. The detection variable of the bus current could easily capture the occurrence of the characteristics change, thus presenting a relatively large value of Q_2. For the arc fault occurring in the resistor branch, i.e. configuration B, it is mixed with the Gaussian noise of the resistor and finally transmitted to the bus. The operation characteristics of masking loads parallel with the arc fault could be more completely transmitted to the bus, so it could reduce the signal to noise ratio (SNR) of the arc fault information, causing difficulties in arc fault detection, thus presenting a relatively low value of Q_2. When the arc fault occurs in the bus position, the arc fault noise is mixed with operation characteristics of masking loads and the resistor noise. In this case, the interference of masking loads to the arc fault detection is between the interference for system configuration B and C. Therefore, system configuration B makes arc fault detection relatively difficult, but the ICEEMDAN based detection variables still works well.

In this section, the ICEEMDAN algorithm introduces an adaptive noise adding process to extract effective information for multi-branch arc fault detection of different masking loads. Two quantitative indexes are constructed to describe the ability of variables to distinguish arc fault from the load adding process. The ICEEMDAN based detection variables are superior to wavelet based detection variables. For different arc fault positions, it is more difficult to detect when the arc fault occurs in the resistor branch. In this case, the proposed detection variable construction method is still effective.

4. Design and simplification of multi-branch arc fault detection algorithm

Arc fault detection requires not only accurate detection of the occurrence of arc faults, but also requires the detection process to be as fast as possible, so as to take protective measures in a short time and reduce the harm caused by the arc fault. The extraction of arc fault detection variables and the design of the detection algorithm need to consider efficiency. By simplifying the two processes respectively, the rapid and accurate detection of arc faults is realized and finally verified well on the hardware.

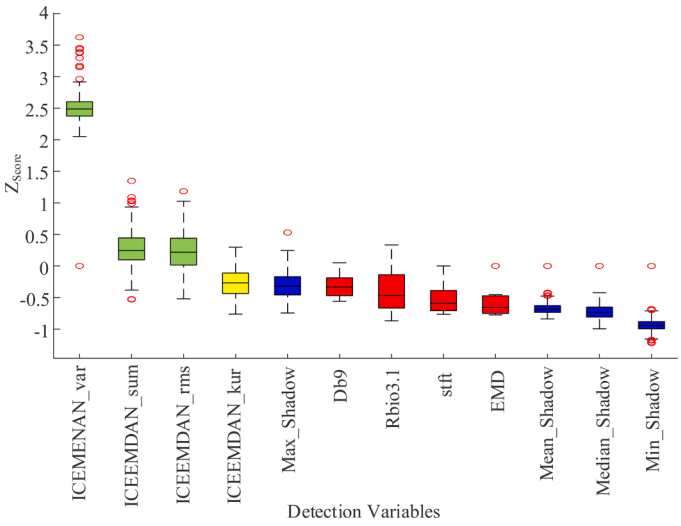


Fig. 11. Mark results of different detection variables obtained by the Boruta algorithm.

4.1. Design of effective feature selection algorithm

It is impossible to obtain the most effective detection variables only by determining the basic construction method. The selection of IMF components and further data processing could extract the arc fault information more fully, which would generate a lot of potential selection features. Some of them includes useful information, others includes redundant and even irrelevant information. Feature selection method could select the most effective feature from the feature pool, thereby reducing the feature dimension and simplifying the calculation process.

The main feature selection methods include filter, wrapper and embedded methods, which have their own advantages and disadvantages. Boruta algorithm, as one of the wrapper feature selection method, was proposed by Kursa et al. in 2010 [33]. It does not pay attention to the importance competition between features, but competes with the original features by constructing shadow features to filter the effective feature set. The importance of features would be given in combination with the random forest algorithm, avoiding the random volatility problem caused by the separate application of the random forest algorithm [34]. Boruta algorithm could also be combined with XGboost and other algorithms to achieve effective feature selection. It could be defined as follows:

Step1: Create shadow features S by randomly shuffle and arrange each original feature R, and form a new feature matrix N=[R, S].
Step2: Input the feature matrix N into the XGboost algorithm, and acquire the importance of each feature in the form of supervised learning. Take the maximum value s_{max} of the importance of shadow features. If the importance of the original feature is greater than s_{max}, it is considered that the feature contributes to the prediction result.
Step3: By iteratively recording the times that each feature is more important than the shadow feature, the accepted important features and rejected unimportant features are finally judged.
Step4: Return the optimal feature subset R_{op} after the iteration. Calculate the score for each feature:

$$MSE_{OOB} = \left(y_i - \hat{y}_{i_{OOB}}\right)^2 \Big/ N \tag{13}$$

$$Z_{Score} = \text{mean}(MSE_{OOB}) / \text{sd}(MSE_{OOB}) \tag{14}$$

where MSE_{OOB} is out of bag error, y_i and y_{iOOB} are the sample value and the predictive value of the out-of-bag sample. Z_{score} is the feature score and sd is the operator to obtain the standard deviation.

For each IMF component obtained by the ICEEMDAN algorithm, in order to further reduce the calculation time, basic time-analysis methods are used to build the detection variable, such as sum, variance, average, kurtosis and root mean square. Wavelet transform, short time Fourier transform, EMD are also used to form the feature pool.

Fig. 11 shows the filtering and marking results of different detection features by the Boruta algorithm. The blue part is the shadow feature, which is meaningless. The red part is the detection variable considered unimportant. The yellow part is the detection variables that cannot be effectively evaluated before the end of iteration. They are not usually used as input features of the final detection algorithm. The green part is the detection variables recommended by the Boruta algorithm, and their importance is always higher than the corresponding shadow feature. The result shows that the Boruta algorithm focuses on the performance of detection variables during the resistor addition process for multi-branch arc fault detection. The detection variables based on the ICEEMDAN algorithm are generally better than other methods such as the wavelet transform, which verifies its applicability and effectiveness.

For each analysis time window, the measured current signal is processed by the ICEEMDAN algorithm first, and then the arc fault detection variables are extracted in the form of variance, sum and root mean square. On the basis of ensuring that the arc fault could be distinguished, the feature dimension is finally reduced as much as possible.

4.2. Design of arc fault detection algorithm based on the lightgbm algorithm

To build a suitable arc fault detection algorithm, it is generally necessary to comprehensively consider the detection accuracy of the algorithm and the complexity of hardware implementation. It is significant to not only accurately detect the occurrence of arc fault and cut off the fault loop in time, but also require that the algorithm is not too complex. In this case, the cost of the designed hardware for the arc fault detection could be controlled.

Arc fault detection algorithms are usually designed based on the threshold comparison method and intelligent methods such as machine learning methods. The threshold comparison method is simple and fast, but its fixed threshold parameters make it difficult to effectively detect arc faults in complex environments with multiple loads. The machine learning method could adapt to the demands of multi-scene arc fault detection by training enough experimental data, and has a high accuracy. However, the overly complex algorithm structure occupies a large amount of memory, which seriously slows down the computing speed, making machine learning difficult to implement in hardware.

The Boosting method in the ensemble learning could continuously adjust the weight of error samples through serial iteration, which could better prevent the over-fitting of the algorithm [35]. It combines many weak classifiers to form a stronger classifier, which could finally achieve accurate prediction classification. The weak classifier type based on decision tree method is often used for the training process of ensemble learning. Gradient boosting decision tree (GBDT) uses gradient descent to optimize in function space, so as to reduce the loss function as quickly as possible. Each iteration produces a weak classifier, which is trained on the basis of the previous classifier residual. Extreme gradient boosting (XGboost) implements the parallel calculation of multithreading on the basis of GBDT [36]. It calculates the loss function with second-order Taylor expansion, which could further improve the accuracy.

XGboost uses the level-wise growth strategy, which could split the same layer of leaves at the same time by traversing the data only once. The strategy makes it easy to optimize with multiple threads, avoiding the over-fitting problem of the algorithm. However, it introduces the problem of low efficiency, which results in poor training and prediction speed. The level-wise strategy treats the same layer of leaves

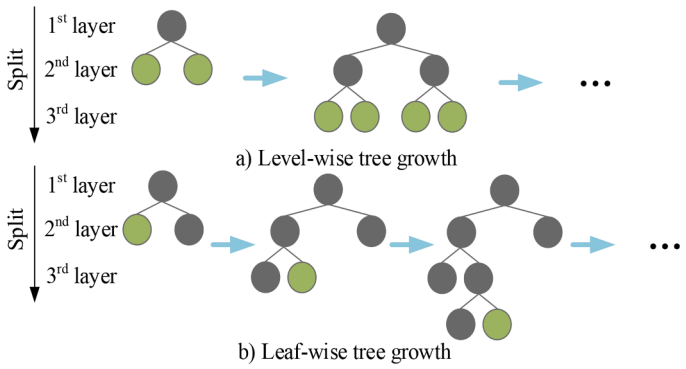


Fig. 12. Comparison of different tree growth modes.

indiscriminately, which is shown in Fig. 12a). In fact, many leaves have a low splitting gain and it is useless for them to continue to be split.

In order to further accelerate the operation speed of the model and reduce the memory occupied by the algorithm, the light gradient boosting machine (LightGBM) algorithm adopts the leaf-wise growth strategy [37]. As shown in Fig. 12b), the strategy finds the leaf with the largest gain of splitting from all leaves in the same layer each time and then make the selected leaf split. Therefore, when the number of splits is the same, that is, the operation time is the same, the leaf-wise strategy could reduce more errors and obtain better prediction accuracy. To prevent over-fitting of the algorithm, it is necessary to control the depth of the decision tree. In addition, LightGBM algorithm uses histogram algorithm to store a large amount of data in the histogram, which further reduces the memory consumption of the algorithm and improves the operation efficiency.

Fig. 13 shows the designed arc fault detection algorithm based on the LightGBM algorithm. The signal is sampled according to the length of the time window. The current data of each time window is analyzed by the ICEEMDAN algorithm to extract detection variables and the detection variable type is pre-selected by the Boruta algorithm. The LightGBM algorithm would output the prediction result of each time window. Output 0 represents that the system is in the normal state and 1 represents the arc fault might been occurred. Comprehensively considering the prediction of the last five time windows, if the number of output 1 is no less than 4, it is considered that the system has already an arc fault. The algorithm would send an alarm prompt and drives the protection device to cut off the fault loop. By judging whether there is output 1 in the last five time windows, it could be determined whether the system state is in the normal state or fault-like condition due to normal current change. The algorithm would return to the current sampling process to continue detecting the arc fault occurrence.

The arc fault detection results for the vacuum cleaner using the proposed detection algorithm on the MATLAB platform is shown in Fig. 14. After the arc fault occurs, outputs of the LightGBM model are all 1, which means the algorithm could distinguish the arc fault well. After the arc fault burns continuously for 87 ms, the alarm requirements of the arc fault detection algorithm are met and the judgment level changes from 0 to 1.

4.3. *Hardware implementation and detection results of the proposed algorithm*

Considering the application of the detection algorithm, the Raspberry pi 4b is used to complete the hardware implementation of the arc fault detection algorithm. With low price and small volume, it could be integrated with circuit breakers and other protective devices, which has sufficient market application potential. Fig. 15 shows the detection variables and results calculated on the Raspberry pi platform for the

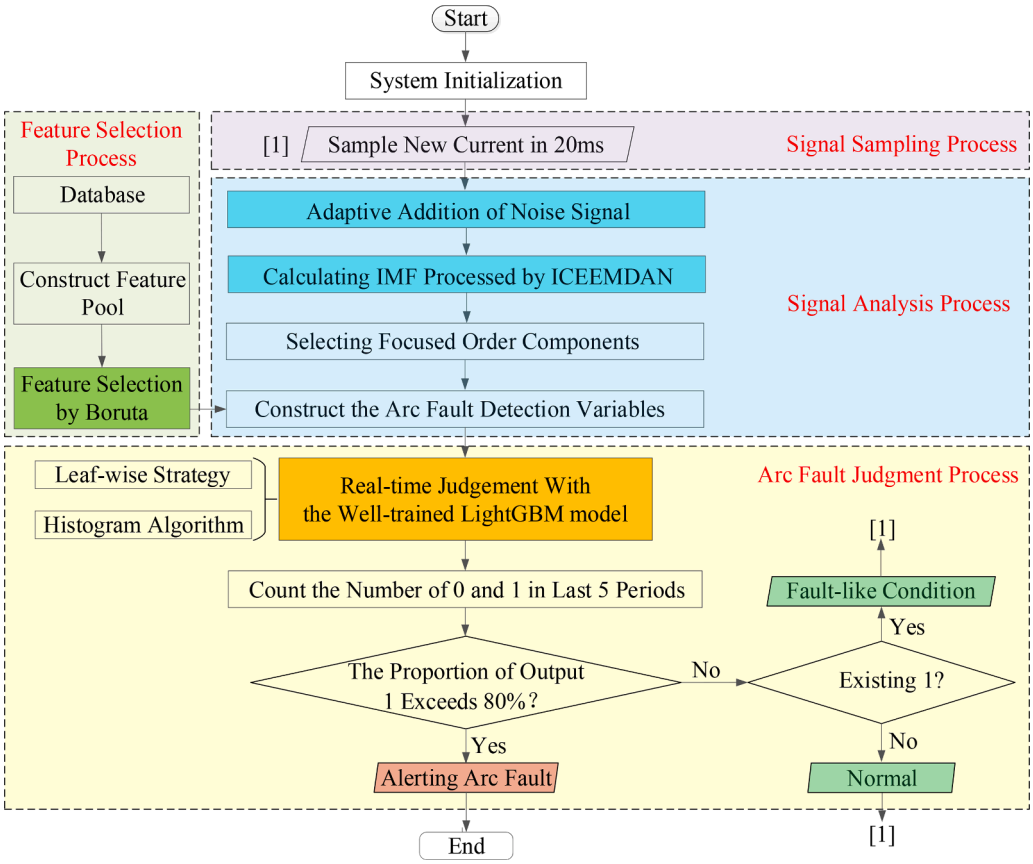


Fig. 13. The arc fault detection algorithm flowchart based on the ICEEMDAN and the LightGBM algorithms.

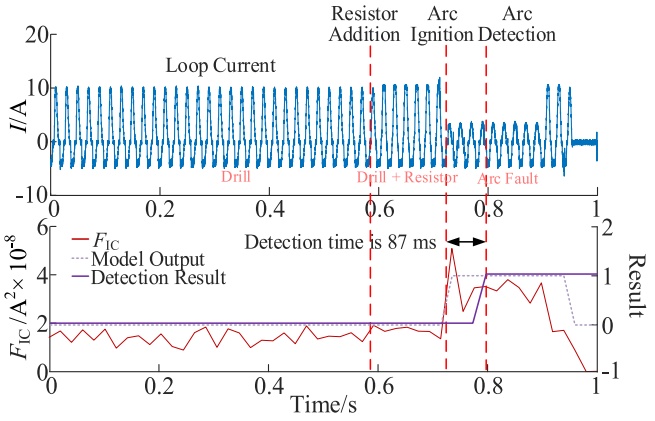


Fig. 14. Arc fault detection results for the vacuum cleaner on the MATLAB platform (multi-branch).

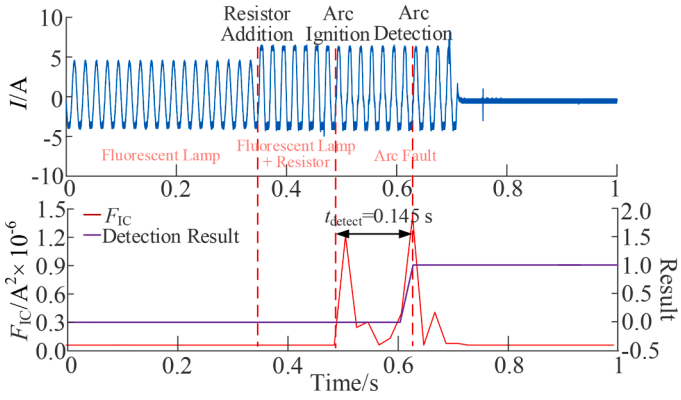


Fig. 15. Arc fault detection results for the fluorescent lamp on the Raspberry pi platform (multi-branch).

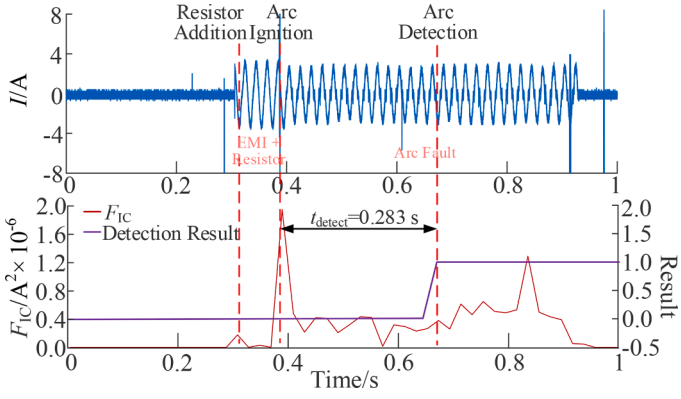


Fig. 16. Arc fault detection results for the resistor with the EMI filter on the Raspberry pi platform (single branch).

fluorescent lamp multi-branch arc fault. The detection variable based on the ICEEMDAN algorithm could distinguish the arc fault from the normal state. The detection time is 0.145 s, which shows that the proposed algorithm could complete the accurate and fast arc fault detection.

In order to better test the applicability of the proposed arc fault detection algorithm, the arc fault detection results of the line impedance and the EMI filter are also considered. As shown in Fig. 16, the arc fault detection result for the resistor with the EMI filter is calculated on the Raspberry pi platform. The detection time is 0.283 s and the hardware application performance is still satisfying.

As shown in the Table 2, the proposed algorithm is compared with other state-of-the-art arc fault detection algorithms. In order to obtain the detection accuracy of the selected algorithms, the experimental data considers 9 types of loads mentioned in Table 1, with a total of 2400 samples of single branch arc faults and 10,300 samples of multi-branch arc faults. The detection accuracy is also calculated separately according to different scenarios. It could be found that the proposed algorithm has great advantages for multi-branch arc fault detection. It has shorter operation time, so it is more suitable for practical application. The detection algorithm is integrated with the circuit breaker to cut off the circuit after detecting the arc fault. By adjusting the algorithm parameters to adapt to different scenarios, the application range of the detection algorithm is improved.

5. Conclusions

Based on the above research, some conclusions could be drawn as following:

1) For multi-branch AC arc fault detection, the load addition process would change the time-frequency characteristics of the current loop, which is easy to lead to the arc fault detection maloperation. The ICEEMDAN algorithm is used to reduce the residual noise in the signal analysis process by adding the Gaussian noise with the adaptive amplitude. The optimal combination of detection variables is constructed and selected by applying the Boruta algorithm. The arc fault detection algorithm based on the LightGBM algorithm realizes the high-precision arc fault detection by constructing a histogram and using the leaf-wise strategy.
2) Two evaluation indexes, Q_1 and Q_2, are constructed to quantitatively evaluate the detection variable performance. $Q_{1,IC}$ is reduced to 0.447 times on average of $Q_{1,WT}$ and $Q_{2,IC}$ is increased to 8.47 times on average of $Q_{2,WT}$. The ability of the ICEEMDAN algorithm is verified to distinguish arc faults from the load addition process. When the arc fault occurs in system configuration B, the extracted arc fault information is less, which is more likely to cause the detection difficulty. In this case, the proposed algorithm could still sufficiently extract the obvious arc fault information.
3) The Boruta algorithm is used to further reduce the feature dimensions, simplifying the calculation of the algorithm. The LightGBM algorithm is used to reduce the memory occupation of the algorithm and improves the operation speed of the algorithm. Based on the Raspberry pi 4b platform, the detection accuracy of the proposed algorithm is 97.06%, and the operation time of a single analysis window is no more than 0.036 s. Finally, an accurate and fast detection algorithm of multi-branch arc faults is realized.

CRediT authorship contribution statement

Yu Meng: Methodology, Data curation, Software, Validation, Writing. **Qi Yang**: Methodology, Investigation, Writing. **Silei Chen**: Writing, Supervision. **Qian Wang**: Conceptualization. **Xingwen Li**: Supervision, Resources, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Table 2
Comparison with State-of-the-art Detection methods.

References	Main detection technology	Number of load type	Sampling frequency	Detection accuracy (single branch)	Detection accuracy (multi-branch)	Maximum operation time for single window
[25]	Wavelet transform + ANN	7	200kHz	81.20%	77.22%	112.09ms
[26]	FFT + SVM	5	2.5kHz	81.12%	74.19%	167.45ms
[38]	VMD, Entropy + SVM	4	409.6kHz	74.00%	72.00%	176.40ms
[27]	CNN - LSTM	6	50kHz	84.08%	79.92%	148.54ms
This Paper	ICEEMDAN + LightGBM	9	100kHz	100%	97.06%	36.00ms

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